Mira Chami

Applied AI Engineer

London, UK · +44 7495 468016 · miranchami@gmail.com · mirachami.dev

Applied AI engineer building LLM systems inside interactive products: grounded generation with citations, knowledge-constrained characters, response verification and model-driven actions in live games. Five years of professional engineering across game engines, services and cross-platform clients.

PROFESSIONAL EXPERIENCE

Tencent

Nov 2023 – Present

Game Tech R&D Engineer (Applied AI & Games) · London, UK

Investigative Game & Applied AI

Gemini, Node.js, Godot

- Built a multi-stage Gemini pipeline with Google Search grounding: source gathering, evidence extraction, witness research and narrative generation as separate stages, each producing structured output validated against JSON Schemas. Stored links come only from pages actually retrieved through grounding.
- Designed AI witness dialogue that separates what a character knows, what they believe and what the player suggests. A separate verification step checks each reply against that knowledge, gives rejected replies one revision, and keeps exact quotes for source-attributed clues.
- Kept question-writing separate from answering: questions are generated only from the evidence the player selected, and the player reviews each one before sending. Owned the Godot case-file, evidence-board and interview UI.
- Made the system reliable and testable: versioned case packages with a comparison viewer for research runs; Node.js/Express services with session restore, caching, validation, timeouts, cancellation and retries; schema, data-integrity and UI tests using mocked model responses.

Extraction Game & Applied AI

Godot, GDScript

- Built the gameplay side of an LLM game director: a validated action library for spawning enemies, rewarding players with items and changing progression, with shared events and entity context as model input. Actions execute only through server-authoritative systems.
- Built a timeline runtime and editor plugin on the same typed actions, so designers can author, preview and validate behaviours by hand while the AI integration develops.
- Architected server-authoritative co-op multiplayer (Steam P2P, LAN, listen and dedicated servers) and data-driven runs, providing the foundation the AI director acts on.

Desktop AI Companion

React, Rust, Tauri

- Built the client side of a game-aware AI companion: prompts submitted with session and game context, HTTP/SSE response streaming with progressive rendering, and persistent per-user chat history.
- Led the Electron vs Tauri evaluation and migration to Tauri v2; built multi-window chat, companion, avatar and settings UI; automated Windows/macOS builds with diagnostic panels and structured logging.

InGame Integration & Multiplayer Demo

UE5, C++, Blueprints

- Built a UE5 multiplayer demo to demonstrate and validate InGame, Tencent’s public online services framework, covering authentication, lobbies, matchmaking, quests, inventory and achievements in gameplay.
- Implemented PlayStation 5 support and dedicated-server debugging; supported partner studios with issues across plugin, backend and UI layers; modified CI pipelines.

Mira Chami

Applied AI Engineer · London, UK · mirachami.dev

PROFESSIONAL EXPERIENCE (CONTINUED)

The Video Games Studio (VGS)

Jun 2020 – Aug 2022

Game Programmer · Beirut, Lebanon

- Shipped two Unity mobile games, owning gameplay architecture, persistence and progression from prototype to release.
- Integrated PlayFab and third-party backend services, and built a separate tool to run live competition cycles and retrieve winner data.
- Built data-driven difficulty, progression and shop content with ScriptableObjects so content could change without code.
- Led programming for a Unity adventure prototype, translating design specifications into character, ability and input systems.
- Supported technical interviews, mentored developers, ran code reviews and maintained gameplay and software architecture documentation.

TECHNICAL TOOLKIT

Applied AI:
LLM Integration · Context Management · LLM Tool Design · Google Search Grounding · Structured Outputs (JSON Schema) · LLM Output Validation · LLM Response Streaming · Source Attribution

Languages:
Python · TypeScript · JavaScript · C++ · C# · Rust · GDScript · Java · C

Backend & Networking:
Node.js · Express · HTTP · SSE · gRPC · Protobuf · InGame · PlayFab · Steamworks · Photon PUN 2

Engines & Desktop:
Unreal Engine 5 · Unity · Godot · Blueprints · React · Tauri · Electron

Build & Engineering:
Git · Perforce · CI · Automated Testing · Mocking · Build & Release Automation · Structured Logging · Performance Profiling

Graphics & Native APIs:
Three.js · PixiJS · Unity URP · Native API Integration

EDUCATION

MSc Games Development (Programming) · Kingston University London · 2022 – 2023

BSc Computer Science · American University of Beirut · 2017 – 2021

BA Economics · American University of Beirut · 2016 – 2020

MISCELLANEOUS

Languages: English, French, Arabic (all native)

Favourite games: Clair Obscur: Expedition 33 · Celeste · Split Fiction · The Last of Us · Outer Wilds